

2.2.2 Smart Supermarket Billing System using RFID

➤ Introduction

The Smart Supermarket Billing System is an RFID-based embedded system developed using Raspberry Pi to automate the billing process in supermarkets. The system scans product RFID tags, displays the selected item details including name, quantity, and cost, and automatically calculates the total amount.

The system also uses a balance card for payment. When the balance card is tapped, the system checks the available balance. If sufficient balance is available, the amount is deducted automatically. If the balance is insufficient, the system displays an “Insufficient Balance” message and allows the user to add a top-up amount. This project demonstrates real-time billing automation using Embedded Systems and RFID technology.

➤ Components Used

Hardware Components:

- Raspberry Pi (Main Controller)
- RFID Reader Module
- RFID Product Tags
- RFID Balance Card
- LCD Display
- Buzzer (Optional for alerts)
- Relay Module (If used)
- Power Supply
- Jumper Wires and Breadboard

Software Used:

- Python Programming
- GPIO Library for Hardware Control

➤ Block Diagram

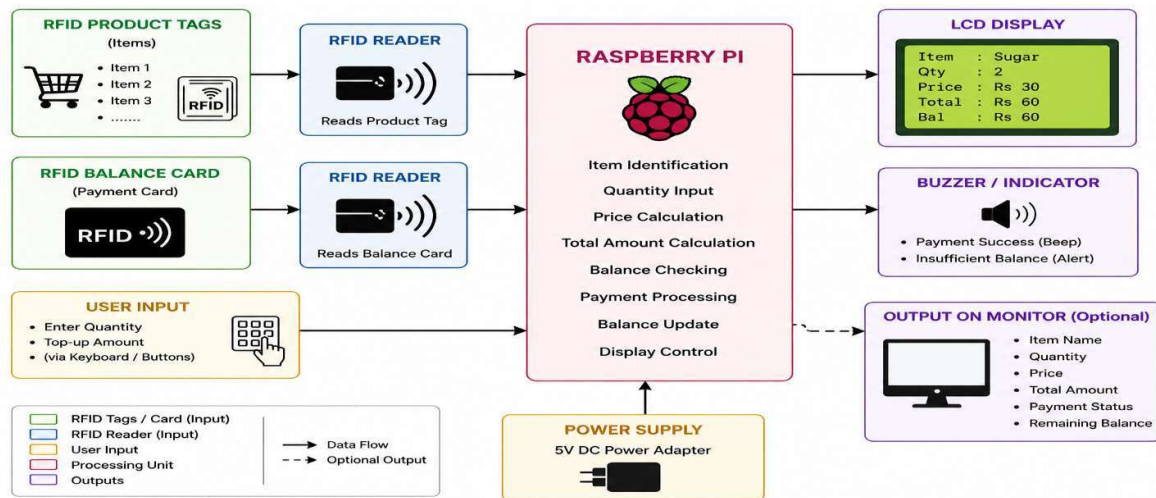


Figure 2.6: Block Diagram of RFID-Based Smart Supermarket System

➤ Hardware Implementation

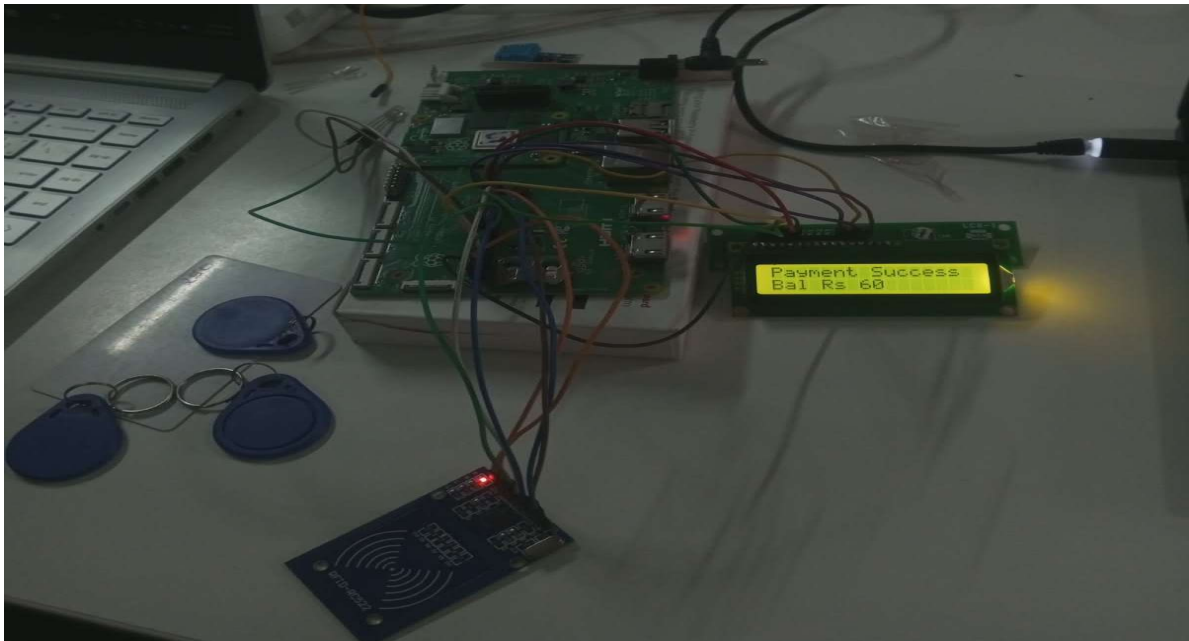


Figure 2.7: Hardware Implementation of the Smart Supermarket Billing System

➤ **Working**

1. The RFID reader scans the product tag when the item is placed near it.
2. The system identifies the item name from the RFID data.
3. The user enters the quantity.
4. The system calculates the total cost based on item price and quantity.
5. The total amount is displayed on the output section (LCD/monitor).
6. The user taps the balance card for payment.
7. The system checks the available balance in the card.
8. If balance is sufficient:
 - The amount is automatically deducted.
 - The remaining balance is updated.
 - “Payment Success” is displayed.
9. If balance is insufficient:
 - The system displays “Insufficient Balance”.
 - The user can add a top-up amount.
 - After top-up, the amount is deducted and updated balance is shown.

This enables a fully automated and cashless supermarket billing system.

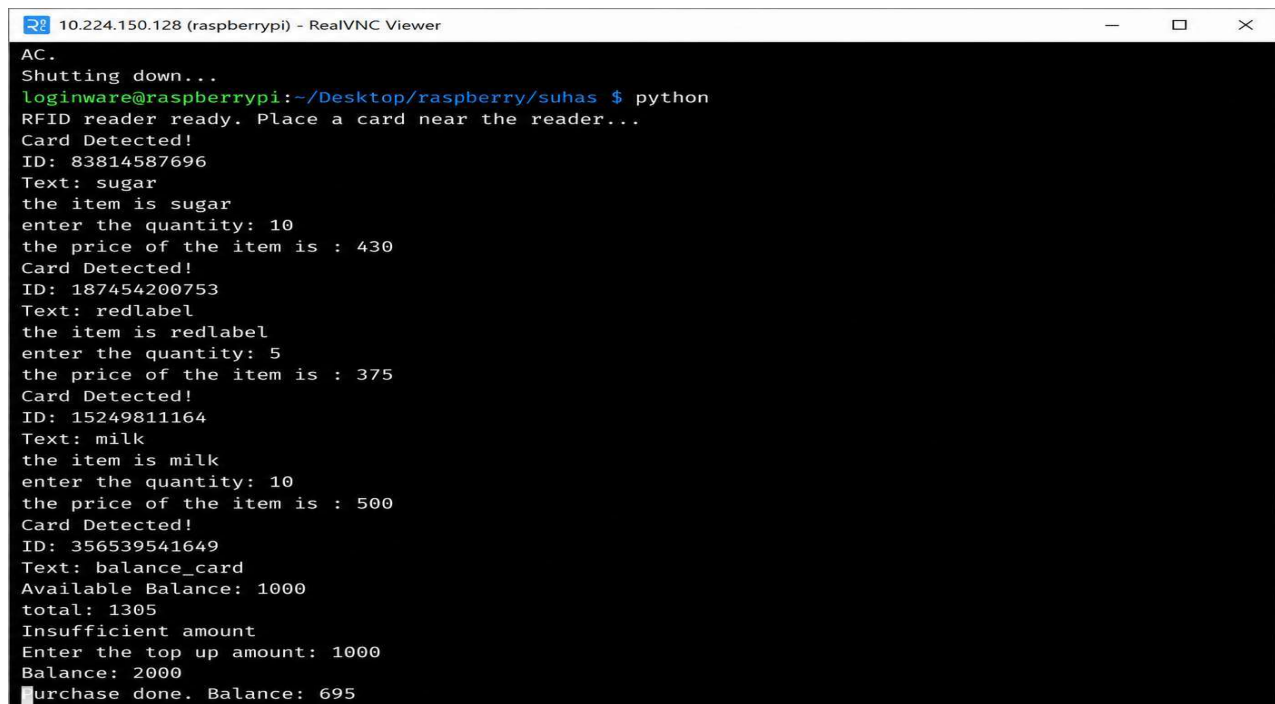
➤ **Results Obtained**

The Smart Supermarket Billing System was successfully tested and operated correctly.

- Items were scanned accurately using RFID tags.
- The system displayed the selected item name, quantity, and price.
- The total cost was calculated automatically.
- The balance card was detected properly.

- If sufficient balance was available, the payment was deducted successfully.
- If balance was insufficient, the system displayed an error message.
- After adding top-up, the balance was updated correctly.
- The final remaining balance was shown on the system.

The project successfully demonstrated automated billing, balance management, and real-time transaction processing using Raspberry Pi and RFID technology.



```

10.224.150.128 (raspberrypi) - RealVNC Viewer
AC.
Shutting down...
loginware@raspberrypi:~/Desktop/raspberry/suhas $ python
RFID reader ready. Place a card near the reader...
Card Detected!
ID: 83814587696
Text: sugar
the item is sugar
enter the quantity: 10
the price of the item is : 430
Card Detected!
ID: 187454200753
Text: redlabel
the item is redlabel
enter the quantity: 5
the price of the item is : 375
Card Detected!
ID: 15249811164
Text: milk
the item is milk
enter the quantity: 10
the price of the item is : 500
Card Detected!
ID: 356539541649
Text: balance_card
Available Balance: 1000
total: 1305
Insufficient amount
Enter the top up amount: 1000
Balance: 2000
Purchase done. Balance: 695
    
```

Figure 2.8: Real-Time Output of the Smart Supermarket Billing System After Hardware Implementation